

Teamwork Agreement for 2026 NACME NVB Engineering Design Project

Project: Data Analytics Apps for a Smart Eco Gym

Team Name: _____

Instructors: Dr. Taylor Lightner, Nicole Jefferson, Lianne R. de Leon

Course: Engineering Design

Members and Roles

(Note: You can add or remove roles depending on team size, and multiple people can serve in multiple roles)

1. _____
2. _____
3. _____
4. _____
5. _____

1) Purpose & Scope

We agree to collaborate to design and deliver a professional-quality data analytics app that transforms raw sensor data into actionable insights for a smart, eco-friendly gym.

Our work will follow the Engineering Design Process and result in:

- A functional MATLAB app
- Well-documented code and datasets
- GitHub repository
- Final presentation

2) Roles & Responsibilities

Each team member will take ownership of a core function. Roles may overlap but all responsibilities must be covered.

- Project Manager (PM)
 - Manages timeline, milestones, and deliverables
 - Facilitates team communication and decision-making
 - Ensures alignment with EDP stages
- Data Engineering Lead
 - Handles sensor data import, cleaning, and preprocessing
 - Documents assumptions, metadata, and data transformation
- Analytics & Modeling Lead
 - Develops metrics, calculations, and algorithms
 - Ensures correctness, units consistency, and engineering reasoning
- App Development Lead
 - Designs and builds the MATLAB app (UI components, interactivity)
 - Connects backend data processing to front-end visualizations
- Validation & Quality Assurance (QA) Lead
 - Test functionality and verifies results
 - Ensures app usability, performance, and accessibility
- Documentation and Communication Lead
 - Maintains README and GitHub repository organization
 - Develops visuals, figures, and presentation materials

3) Communication Norms

- Primary channel: _____ (e.g. Slack)
- Secondary channel: _____ (e.g. E-mail for summaries/decisions)
- Response time: _____ on weekdays and _____ on weekends (e.g. within 24 hours on weekdays; within 48 hours on weekends)
- Meeting cadence: _____ (e.g. weekly working meeting (60–90 minutes on Tuesdays via Zoom))
- Decision records: _____ (e.g. PM logs major decisions in a running document)
- Expectations
 - Come to meetings prepared with updates and blockers
 - Share progress regularly
 - Document major decisions in shared space

4) Collaboration & Tools

- Version control: _____ (e.g. one shared GitHub repo or Google Drive folder)
- Data management: All datasets, parameters, and assumptions documented in _____ (e.g. README.md)
- File naming & style: _____ (e.g. Snake_case, descriptive names)
- Backups: _____ (e.g. PM tags old versions according to numbered system)

5) Quality Standards (Definition of 'Done')

(Note: You can edit, add, or remove the suggestions from this list based on your team discussion)

A task is considered complete when:

- Code runs without errors and produces expected outputs
- Figure and results are reproducible when regenerated
- Assumptions, units, and methods are documented
- App functionality is tested by at least one other teammate
- Work meets project requirements
- _____
- _____
- _____

6) Timeline & Milestones

(Note: You should add specific tasks each team member agrees to complete each week, mapped onto to the course deliverables outlined in the syllabus)

Week 1: Discovering the Problem

Week 2: Designing the Solution

Week 3: Architectural Foundations

Week 4: Building & Qualifying Quality

Week 5: Releasing and Maintaining Impact

7) Decision-Making

- Default: _____ (e.g. consensus after discussion; PM facilitates)
- If blocked >48 hrs: _____ (e.g. Time-limited debate, majority vote, PM documents decision)
- Technical disagreements: _____ (e.g. Evidence-based comparison before vote)
- Escalation: If unresolved, consult instructor/mentor with a concise memo.

8) Accountability

- Task board: _____ (e.g. To-Do / Doing / Review / Done with owners and due dates documented in README.md)
- Accountability: _____ (e.g. Missed deadline → recovery plan within 24 hrs; repeated misses trigger role redistribution.)
- Attendance: _____ (e.g. If you'll miss a meeting, give 24-hour notice and share updates asynchronously.)

9) Inclusion

We commit to:

- Respectful and inclusive collaboration
- Equitable participation and workload
- Supporting teammates with different experience levels

- Flexibility for documented accessibility needs

10) Deliverables Checklist

(Note: You can edit, add, or remove the suggestions from this list based on your team discussion)

- ☐ MATLAB app (.mlapp)
- ☐ Documented code and repo
- ☐ README instructions
- ☐ Visualizations
- ☐ Final presentation
- ☐ _____
- ☐ _____
- ☐ _____

11) Agreement and Signatures

By signing below, we agree to uphold this teamwork agreement.

Name & Signature: _____ Date: _____

Name & Signature: _____ Date: _____

Name & Signature: _____ Date: _____

Name & Signature: _____ Date: _____

Name & Signature: _____ Date: _____